HPC

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Experiment-05

- AIM: Implement HPC Application for AI/ML domain.

- THEORY:

(1). Introduction.

- High performance computing (HPC) referes to the use of parallel processing techniques, advanced hardware architectures and optimized software frameworks. to solve Computationally intensive problems efficiently.
- In recent years, the rapid growth of artificial intelligence and machine learning has significantly increased the demand for HPC systems as modern AI/ML models involve large datasets, high dimensional computations and iterative training processes.

- AI/ML applications such as deep learning model training image and speech recognition, natural language processing and predictive analytics require massive computational power and memory bandwidth.
- HPC provides the necessary infrastructure by leveraging multicore CPUs, GPUs, distributed clusters and high speed interconnects, enabling scalable and accelerated AI/ML workloads.

(2). Relationship between HPC and AI/ML -

- AI/ML algorithms are inherently compute / intensive and data parallel, making them well suited for HPC environments.

- Why AI/ML needs HPC - ?

- Large scale datasets (Big data)
- Complex mathematical operations (matrix multiplications, convolutions).
- Iterative optimization (gradient descent, backpropagation)
- Real time or near real time inference requirements.

* HPC enables -

- Faster model training.
- Efficient handling of large datasets.
- Scalability across multiple notes.
- Reduced time to solution.

(3) Typical HPC architecture for AI/ML applications -

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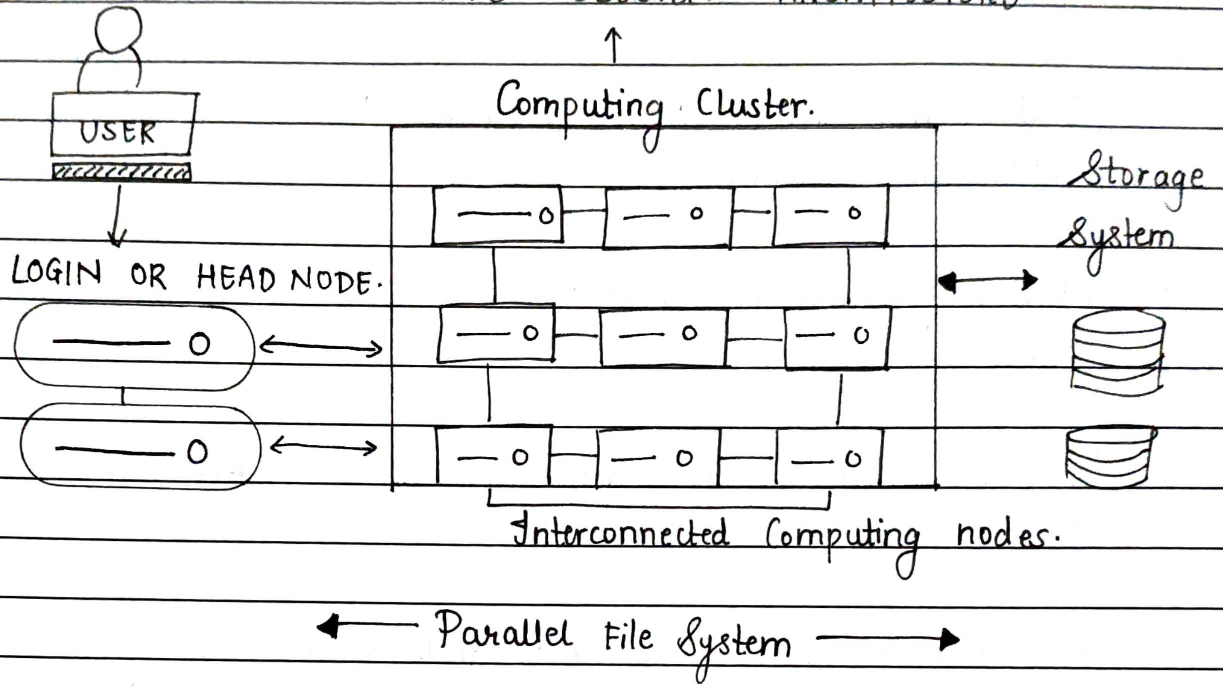
HPC CLUSTER ARCHITECTURE.



(4) Hardware Components -

- Compute Nodes : Multicore CPUs and accelerators (GPU/TPU)
- Memory Systems : High bandwidth RAM and GPU memory
- Storage : Parallel file systems (Lustre, GPFS)
- Interconnects : High speed networks (InfiniBand).

(5) Software Stack -

- Operating system : Linux - Based
- Parallel programming models
  - MPI (distributed memory)
  - OpenMP (shared memory)
  - CUDA / Open CL (GPU acceleration).

- AI/ML frameworks - Tensorflow, PyTorch, Scikit learn.
- Job schedulers - SLURM, PBS.

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(6). Step by Step workflow -

1. Data ingestion - Large datasets are loaded from distributed storage.

2. Data pre-processing - Parallel data cleaning, normalization and augmentation.

3. Model initialization - Weights and parameters initialized across nodes / GPUs.

4. Parallel Training - Forward and backword propagation executed in parallel.

5. Model Synchronization - Gradients aggregated using collective communication.

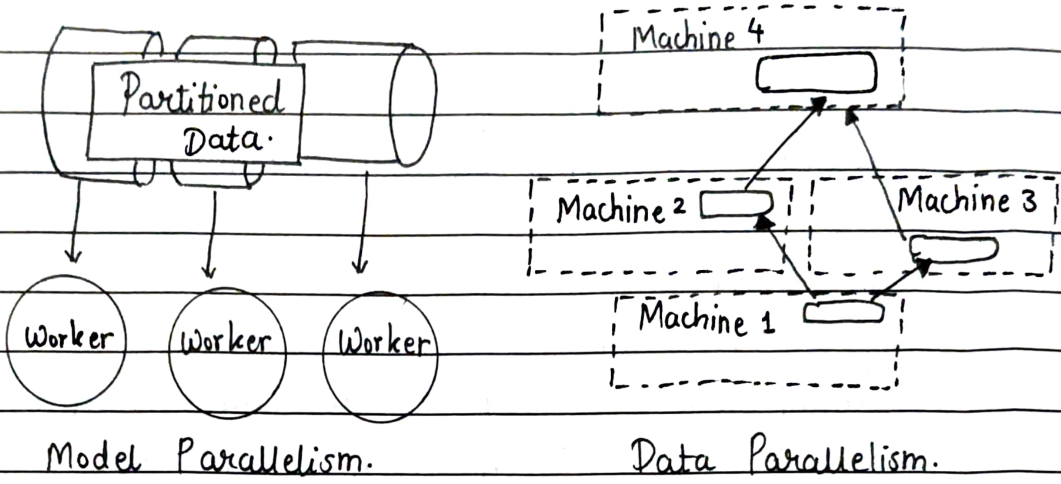
6. Evaluation and inference - Trained model evaluated on test data.

7. Result storage - Outputs stored back into HPC storage systems.

(7) Computational characteristics -

- Large matrix multiplications.
- Repeated convolusion operations.
- Iterative optimization using gradient descent.
- High memory and compute intensity.

(8) Parallelization Strategy -



(a) Data Parallelism -

- Dataset is divided across multiple GPUs or nodes
- Each processor trains a copy of the model.
- Gradients are synchronized after each iteration.

(b) Model Parallelism -

- Different layers of neural network run on different processors.
- Useful for very large models that do not fit in a single GPU.

(9) Advantages of using HPC for AI/ML -

- Massive reduction in training time.
- Ability to handle extremely large datasets.
- Improved scalability and reliability.
- Support for realtime and near real time AI applications.

- Enables complex models like transformers and large CNNs.

(10). Challenges in HPC based AI/ML systems -

- High infrastructure cost.
- Complex programming and debugging.
- Communication overhead in distributed systems.
- Load balancing in heterogenous hardware.
- Energy consumption.

(11). Applications -

- Medical image analysis.
- Autonomous Vehicles.
- Climate and weather prediction.
- Fraud detection.

CONCLUSION -

Thus we studied and implemented application of AI/ML using HPC.